Chapter 1 – Foundations of Digital Forensics

Assignment

Total Questions: 15 | Total Marks: 20

SUBMISSION DEADLINE

28TH August, 2026

Instructions:

- Answer **ALL 15** questions
- Provide detailed, well-structured answers with examples where indicated
- Submit in hard copy
- Plagiarism-free answers in your own words

ASSIGNMENT QUESTIONS

Question 1: Define Digital Forensics. How is it different from routine IT troubleshooting?

Question 2: List and explain the major objectives of Digital Forensics. Why is "discovering the truth" not enough—why must evidence also be legally admissible?

Question 3: Explain the importance of Digital Forensics in modern cybersecurity. Provide any TWO real-world applications mentioned in the chapter.

Question 4: Digital evidence differs significantly from physical evidence. Explain the key characteristics of digital evidence. Also, state why even a simple action like opening a file can compromise evidence.

Question 5: Differentiate between Volatile and Non-Volatile evidence. Give at least TWO examples of each type.

Question 6: List the common sources of digital evidence. Mention any FOUR sources and explain what type of evidence each can provide.

Question 7: Compare and contrast Computer Forensics and Network Forensics using the following parameters:

- Definition
- Primary evidence sources
- Type of evidence (volatile/non-volatile)
- Main objectives
- Common tools

Question 8: What is Mobile Forensics? List any FIVE types of evidence that can be recovered from a smartphone during an investigation.

Question 9: Describe the complete Digital Forensic Investigation Process in detail. For each of the six phases, explain:

- The objective of that phase
- Key activities performed
- Tools or techniques used (if applicable)

Question 10: What is Chain of Custody? Why is it critical in digital forensic investigations? Provide a practical example of how a chain of custody record should be maintained.

Question 11: Explain why cryptographic hashing (e.g., SHA-256) is used during evidence acquisition. What happens if the hash value changes during the investigation?

Question 12 – Scenario Based

Scenario: A multinational manufacturing company discovers that an employee's credentials were used to access design documents from an unusual geographic location (Russia) at 2:00 AM local time. The employee's laptop was shut down at that time.

As an investigator, answer the following:

1. What are the SIX fundamental questions (Who, What, When, Where, How, Why) this investigation must answer? (1 Mark)
2. Which specific evidence sources would you examine to determine if the account was compromised rather than the employee being the perpetrator? (2 Marks)

Question 13 – Scenario Based

Scenario: A ransomware attack has encrypted critical files on a hospital's file server. The initial infection is traced back to an email attachment opened by an employee.

As a Digital Forensic Investigator:

1. List the potential evidence sources you would identify. (1 Mark)
2. Explain how you would preserve the evidence to ensure integrity. (1 Mark)
3. What artifacts would you examine to reconstruct how the attack occurred? (1 Mark)

Question 14: What are the technical and legal challenges faced by digital forensic investigators? Give TWO examples of each.

Question 15: Explain the evolution of cybercrime from the 1980s to the present day. How has Digital Forensics evolved to keep pace with modern threats like Ransomware-as-a-Service (RaaS) and AI-assisted phishing?
